OpenSCAD - Hand Fan Generator Prompt

Create an Open SCAD script that generates handheld personal fan blades that can later be assembled into a personal fan.

CUSTOMIZABLE PARAMETERS

- Number of Blades - the total number of blades that make up the entire fan (default 19)
- Blade Length - the total length of a fan blade (default 180)
- Blade Thickness - the thickness of the fan blade (default 1)
- Top Blade Width - the width of the top or far end of the blade (default 30)
- Bottom Blade Width - the width of the bottom or hand held portion of the blade (default 16)
- Blade Cutouts - The blade can have a pattern cut out of it. (default None) The supported options are as follow:
  - None - no cutout, the blade is solid
  - Voronoi - A voronoi pattern is cut all the way through the thickness of each blade. The walls between the holes should be at least 2mm in thickness but can vary
  - Honeycomb - a honeycomb pattern is cut all the way through the thickness of each blade
  - Hearts - a number of small hearts of varying sizes (within 2mm of each other in size difference) and of varying flat (y-plane) rotation angles cut all the way through the thickness of each blade.
  - Vertical Text - A specific line of text is etched into each blade along the y-axis direction centered between the bottom of the blade to the top of the blade. The etching goes through Blade Cutout Text Depth of the thickness of the blade

    - Horizontal Text - (this only renders when all blades are rendered either for printing or in assembly view. It is ignored when only one blade is rendered in the Rendering mode) A specific line of text that is etched into each of the blades so that when the blades are extended fully around the pivot point and as far as each Connection Slot and Connection Slot Peg allow, the line of text is displayed. The etching goes through Blade Cutout Text Depth of the thickness of the blade. It is OK for a portion of the same character to exist on the next blade if a portion of that character on the prior blade is going to be covered up. Space the duplicate character and following characters appropriately so that they do overlap in the right places when the fan is extended.
- Blade Cutout Text Font - the font to use for any Vertical or Horizontal text cutouts
- Blade Cutout Text Size - the font size to use for any Vertical or Horizontal text cutouts
- Blade Cutout Text Depth - the depth of the cutout of the text for any Vertical or Horizontal text cutouts
- Blade Cutout Text Color - (default: yellow) - The color of the etched text for any Vertical or Horizontal text cutouts
- Blade Cutout Vertical Text (default "") - the text for the Vertical Text cutouts
- Blade Cutout Horizontal Text Line 1 (default "") - The text for the Horizontal Text cutouts across the blades.
- Blade Cutout Horizontal Text Line 2 (default "") - The text for the second line (if any) of Horizontal Text cutouts across the blades. If it is not populated, Line 1 is centered vertically across the blades without calculating the space that would have been taken by Line 2. If Line 2 is populated then the area of Line 1 and Line 2 is centered vertically on the blades.
- Rendering Mode - What blades will be generated and in what orientation
    - All Blades for Printing - all of the blades are generated and laid next to each other however they should be rotated around the y-axis by 180 degrees so that they are alternating orientation to fill up the space better. Blades should be alternating between Blade Section A and Blade Section B. The bottom connector cap cylinder and it's associated cap should be generated as well.
    - Blade Section A - just one blade and it should be the blade section where the top/end blade connector is in the top slot on the right end of the slot.
    - Blade Section B - just one blad and it should be the blade section where the top/end blade connector is in the bottom slot on the right end of the slot

- Assembly View - all of the blades stacked on each other and spread out as if they were assembled. They should be fixed together at the bottom connector hole and then their top ends rotated to the right of the previous blade by the full width of the blade minus the connection slot margin width and the connection slot connector width. Blades should be alternating between Blade Section A and Blade Section B.

- Blade Bottom Connection Hole Diameter (default 4mm)
- Blade Bottom Connection Hole Bottom Margin (default 24mm) - The distance between the bottom of the Bottom Connection Hole and the bottom of the blade
- Blade Bottom Connection Peg Tolerance (default 0.2) - The spacing between the outer edge of the Peg and the Connection Hole
- Blade Bottom Connection Peg Cap Tolerance (default 1mm) - the spacing between the bottom of the Peg Cap and the stack of blades that have been slid into the connector hole.
- Blade Bottom Connection Peg Base Diameter (default 8mm)
- Blade Bottom Connection Peg Base Thickness (default 2mm) - The distance between the bottom of the peg base and the top of the peg base before the connection shaft is added to it.
- Blade Bottom Connection Peg Cap Diameter (default 8mm)
- Blade Bottom Connection Peg Cap Thickness (default 4mm) - the distance between the bottom of the peg cap and the top of the peg cap without the cylindrical cutout for the actual peg.
- Connection Slot Peg Tolerance (default 0.3) - the tolerance level that extends from the thickness of the blades to the triangular extension of the Connection Slot Peg.

HIDDEN PARAMETERS

- Blade Cutout Top Margin: default 25 - This is how far from the top end of the blade before the cutout pattern ends
- Blade Cutout Bottom Margin: default 80 - This is how from from the bottom end of the blade before the cutout pattern can start
- Blade Cutout Side Margins: default 5mm - This is how from from the blade edges that cutout pattern should run up to but not exceed.
- Blade Cutout Pattern Inner Wall Thickness: default 2 - The walls between the holes should be at least 2mm in thickness but can vary depending on the type of cutout pattern selected
- Blade Connection Slot Side Margin: default 5mm - This is the margin between the edge of the connector slots and the end of the blade
- Blade Connection Slot Top Margin: default 10 - This is the margin between the top end of the blade and the top edge of the top connection slot.

- Blade Connection Slot Inner Margin: default 10 - This is the margin between the bottom of the top connection slot and the top of the bottom connection slot.
- Blade Connection Slot Height: default: 6 - this is the total height of each individual blade connection slot
- Blade Connection Slot Lip Width: default 1.6 - This is the number of mm where the blade connection clot lip extends into the Blade Connection Slot area from both the top of the slot and from the bottom of the slot.
- Blade Connection Slot Peg Tolerance: default 0.2 - This is the tolerance gap between the top and bottom edges of the blade connection slot peg and the edges of the Blade Connection Slot Lip edges.
- Blade Connection Slot Peg Width (default 3mm) - This is the width of the peg itself along the x-axis assuming the blade is oriented so that the sides of the blade run up and down the y-axis and the distance between the sides of the blade or measured along the x-axis.

BLADE COMPONENTS

There are two types of blades that make up the fan. These are blade A and blade B. Blade A and blade B will be alternated as they are connected together to make up the fan. there is only one slight difference between each blade, and that is in the location of the blade connection slot peg. On blade A, the peg will be in the top connection slot at the far-right end. For blade B, the blade connection slot peg is in the bottom slot on the far-right of the slot

A sample STL file of Blade A and Blade B can be found in files: "Hand Fan Sample Blade A.stl" and "Hand Fan Sample Blade B.stl" respectively.

There are two connection mechanisms on a blade: the bottom connection point and the top connector slots.

Bottom Connection Point

The bottom connection point is a hole in the blade near the bottom of the blade. It is through this hole that the Blade Connector Peg will be inserted to be the primary pivot point for rotating out the fan blades. It consists of tow parts, the Peg and the Peg Cap. The Peg has a cylindrical base and then it has a cylindrical rode that extends from it. The rod length should equal the (blade thickness * number of blades + Connection Peg Cap Thickness + Connection Peg Cap Tolerance). The cap should be a cylindrical object with a hole cut into it that is (Connection Peg Cap Thickness - Connection Peg Cap Tolerance) deep.

A sample STL file of the peg exists in "Hand Fan Sample Connector Peg.stl"
A sample STL file of the peg cap exists in "Hand Fan Sample Connector Peg Cap.stl"

Top Connector Slots

These consist of two slots near the top of the blade. Distances and margins are defined in the Parameters for the model. A slot consists of a rectangular cutout in the blade that has a bottom lip on the top and bottom of the slot. The lip allows a slot connector to slide across the slot when expanding the fan blades.

The Connection Slot Peg is trapedoizal object that fills in and extends from right side of one of the slots. It has a rectangular base that extends beyond the top of the blade by the thickness of the blade. It's length is the is the distance between the edges of the two lips in the slot plus the Connection Slot Peg Tolerance. From there it extends at an angle until the top of the connector is as wide as the full Connection Slot minus 1 mm.